

Chemistry — Assignment 6: Counting σ and π Bonds

Anshuman Atrey · 150096724029 · BTech CSE 2024-28

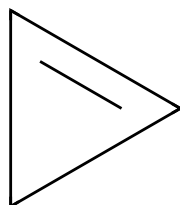
Chemistry (Foundational Chemistry) · Semester 5

Question. Find the number of sigma and pi bonds in: (i) ethyne, (ii) cyclopropene, (iii) tetracyanoethylene, (iv) 1,3,5-trimethylenecyclohexane. Also: count the pi bonds in a given compound.

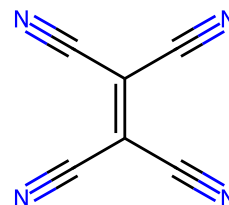
Rule. single bond = 1σ ; double = $1\sigma + 1\pi$; triple = $1\sigma + 2\pi$. Count every bond, including each C–H. (σ/π below computed from each structure.)



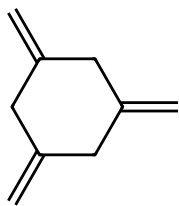
ethyne



cyclopropene



tetracyanoethylene



1,3,5-trimethylenecyclohexane

Compound	formula	σ bonds	π bonds
ethyne	C ₂ H ₂	3	2
cyclopropene	C ₃ H ₄	7	1
tetracyanoethylene	C ₆ N ₄	9	9
1,3,5-trimethylenecyclohexane	C ₉ H ₁₂	21	3

Counting π bonds (Q2)

π bonds = number of double bonds + $2 \times$ (triple bonds). E.g. benzene = **3 π** , naphthalene = **5 π** , anthracene = **7 π** , CO₂ (O=C=O) = **2 π** .